Segmenting Metastatic Prostate Cancer in Whole-Body PSMA PET Scans with a Hybrid CNN-Mamba Network

James Wigfield

Supervised by Dr Jake Kendrick and Dr Mubashar Hassan

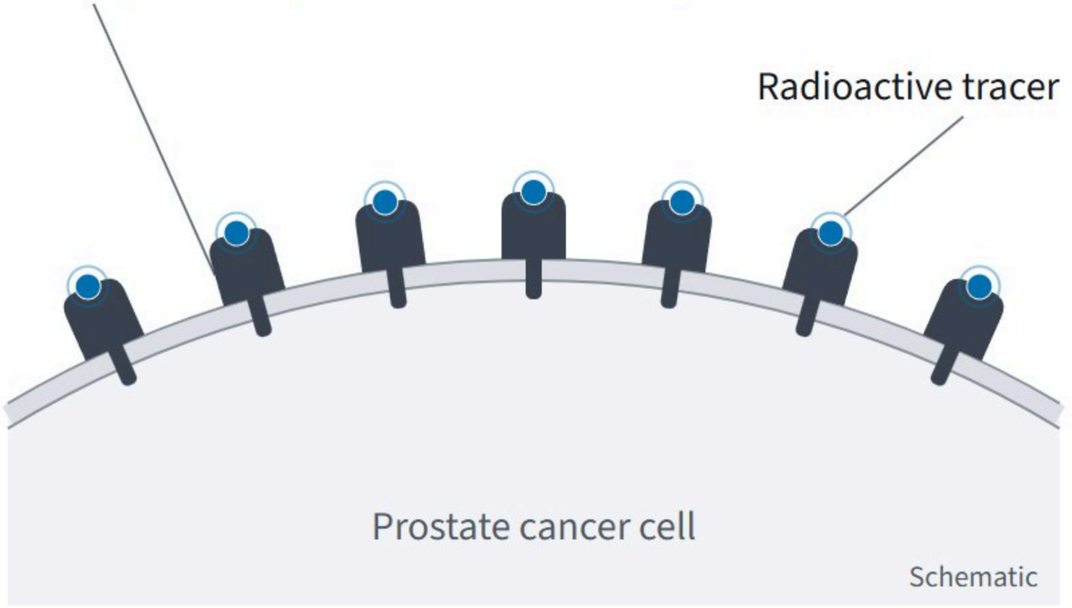


Metastatic Prostate Cancer and PSMA PET/CT

35% or less

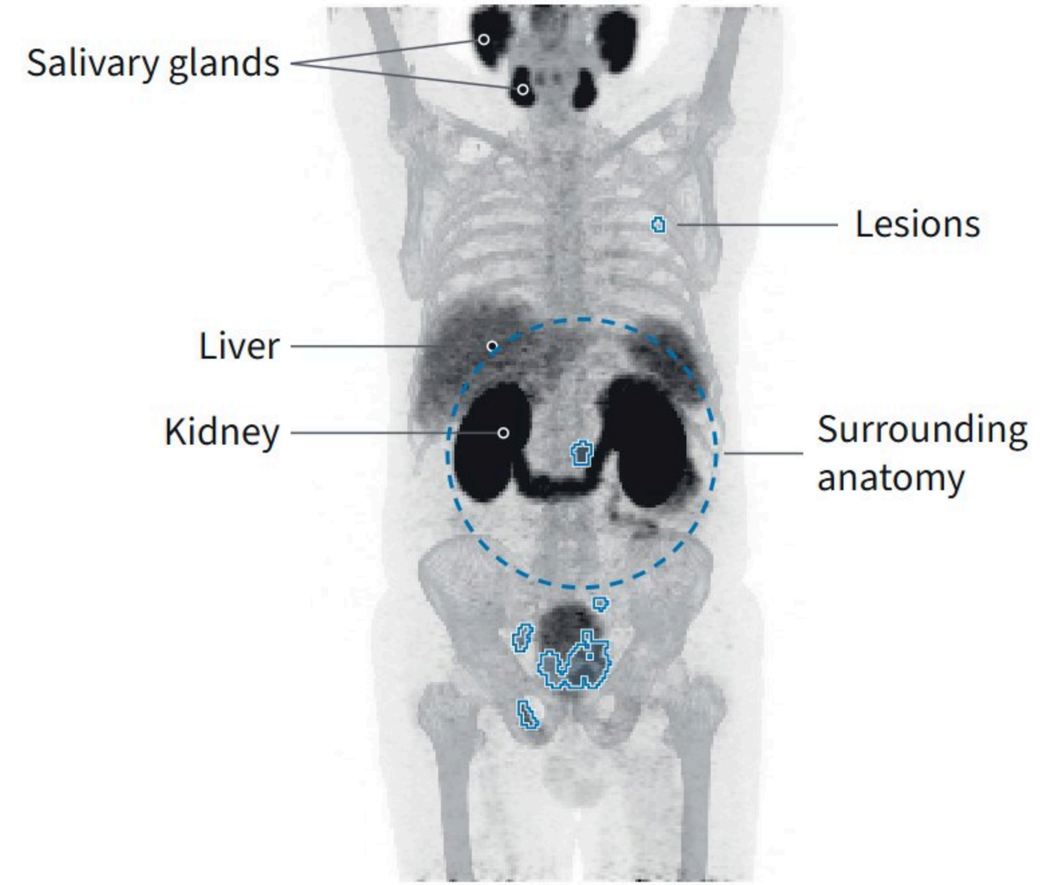
five-year survival once it spreads

PSMA (prostate-specific membrane antigen)



NORMAL UPTAKE

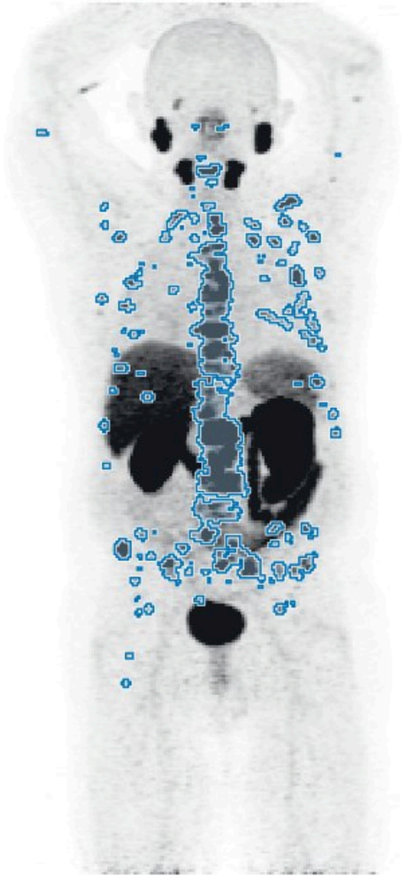
PSMA PET/CT



[1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022 · [2] Sheikhbahaei et al., Eur J Nucl Med Mol Imaging 2017

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Manual Outlining and Expert Agreement

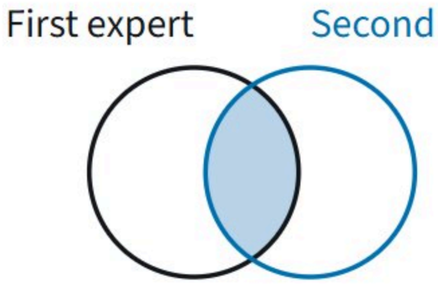


One scan from this project's data

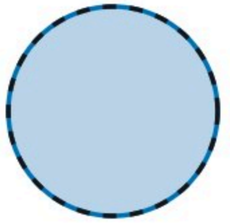
Over 100 lesions
in some scans

2022 study, Kendrick et al.

Two experts, the same 28 scans



33%
Dice, on average



100%
Identical outlines

Second expert found about 7 out of 10



[1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

PSMASegmentator and the Patch Size Limit

Open-source tool by Joel Noble, PhD student,
UWA Medical Physics Research Group

Built on nnU-Net, a CNN framework
that configures itself to any new dataset

Trained on 1,035 cases

3D CNNs can't see the entire scan at once

GPU memory limits patch size

☐ PSMASegmentator's patch, to scale: 288 x 128 x 128 voxels

Coronal view; front to back not shown



[3] Isensee et al., Nature Methods 2021

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Mamba, a Selective State Space Model

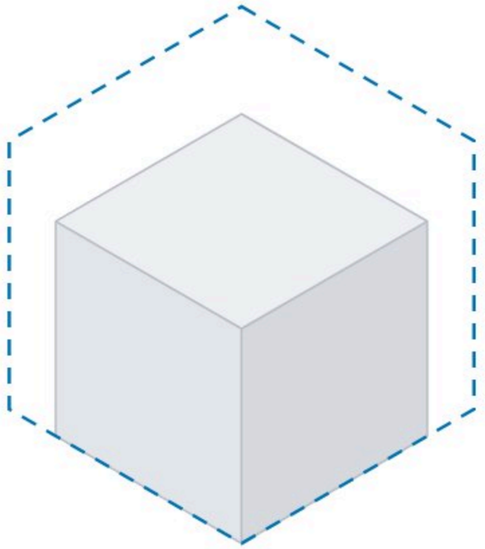
One step at a time, with a fixed-size memory

Selective: kept ignored



Flattened into a long 1D sequence

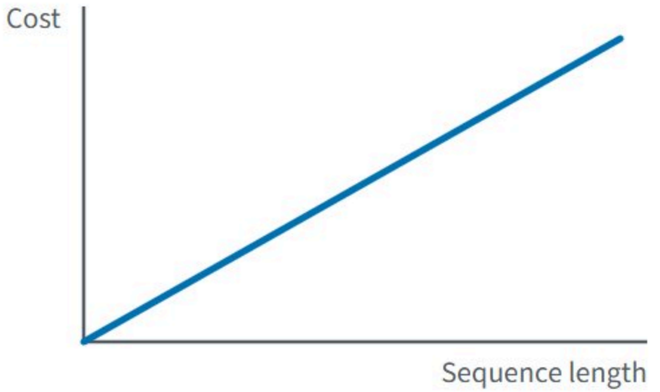
Views across the whole patch



3D image features

Could free memory for a bigger patch

Cost grows linearly with sequence length



Schematic

[4] Gu and Dao, arXiv 2023

Research Question

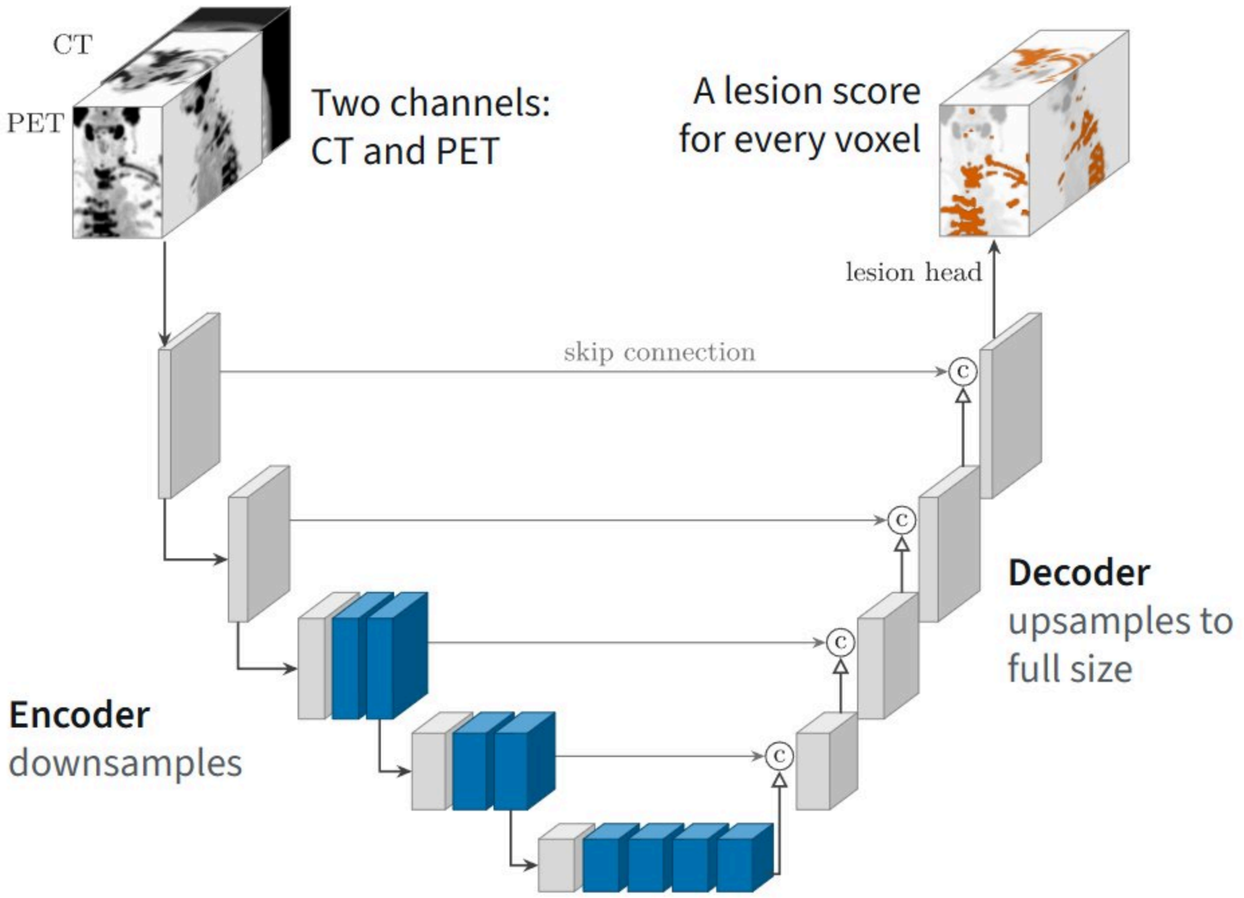
Hybrid CNN-Mamba vs PSMASegmentator

Does a hybrid CNN-Mamba network find lesions more effectively than PSMASegmentator on scans neither was trained on?

Hybrid CNN-Mamba vs CNN-only version

Do the Mamba blocks improve lesion detection, compared with a CNN-only version of the same network?

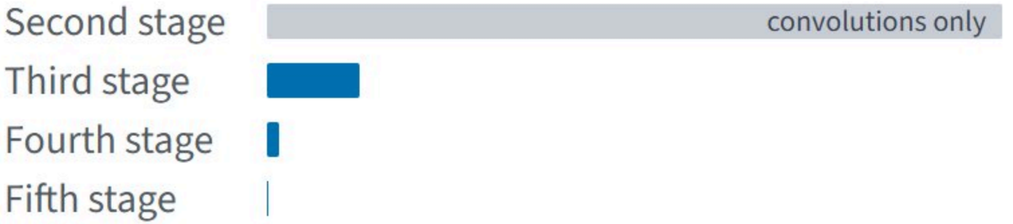
Hybrid CNN-Mamba Network Architecture



Convolutions in every stage

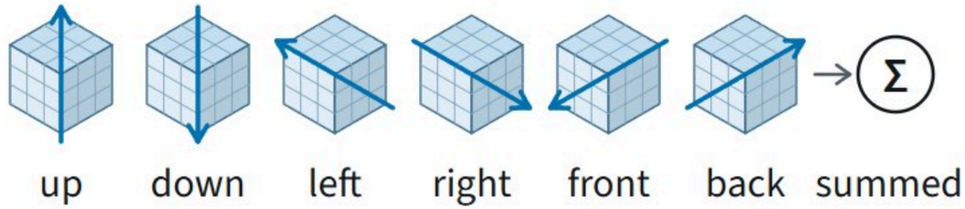
Mamba blocks only in the three deepest stages

Flattened sequence length, to scale



About eight times shorter at each stage

Each Mamba block: six directions, summed



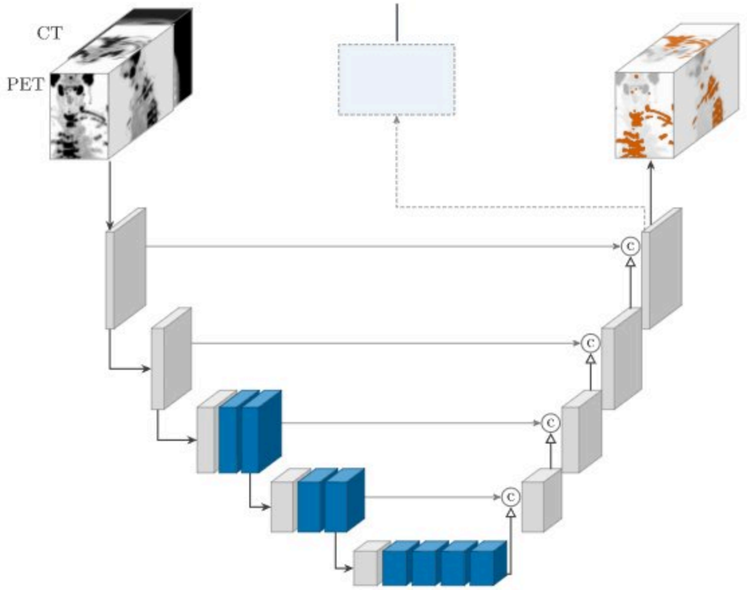
Schematic

[5] Xing et al., IEEE Trans Med Imaging 2026 · [6] Xing et al., MICCAI 2024

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Organ Head and Five-Model Ensemble

Second output, used during training only
outlines 24 anatomical structures



Earlier version

Flagged lesions in all seven
scans that had no lesions

●●●●●●●○○○○○○○○○○○○○○

21 → 7

before — with the organ head

false alarms on those scans
with nothing else changed

The final model on one test scan



Final model: five versions



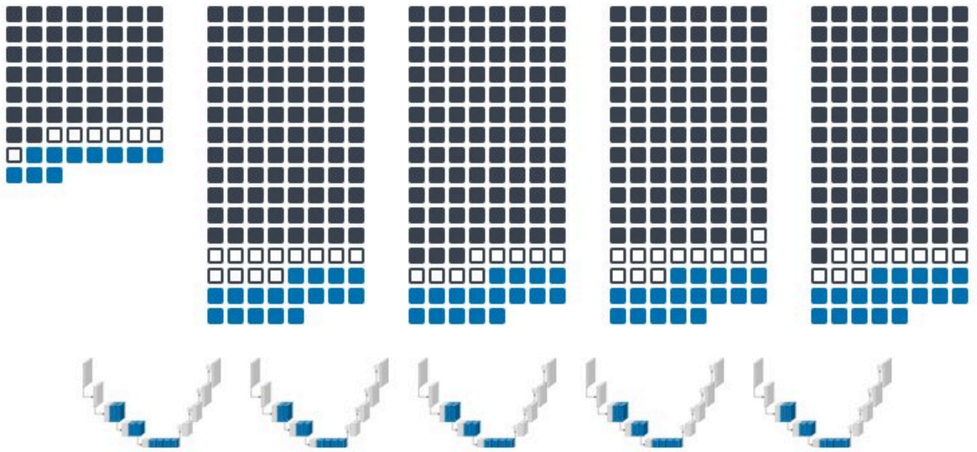
each trained on a different portion of the data
averaged voxel by voxel; threshold 0.5
predicted lesions under 10 voxels removed

[7] Rokuss et al., arXiv 2024 · [8] Wasserthal et al., Radiol Artif Intell 2023 · [2] Sheikhbahaei et al., Eur J Nucl Med Mol Imaging 2017
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

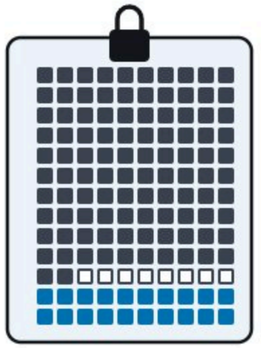
Datasets, Test Set and Scoring

- **autoPET-III:** 597 scans, ▢ 58 with no lesions
  labels: SUV threshold, then corrected by an expert
- **DEEP-PSMA:** 100 scans, advanced disease, every scan has lesions
  labels: SUV threshold, normal uptake removed by a physician

The other 567: five-fold cross-validation



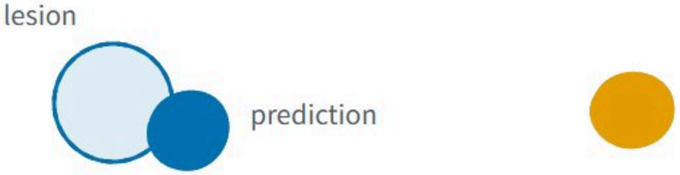
the five versions in the ensemble



130 sealed away
never trained on, scored once

Scoring

My model and **PSMASegmentator** used the same scoring script
written by Joel Noble, used with his permission



Found: a lesion that any predicted voxel touches

False alarm: a prediction touching no lesion

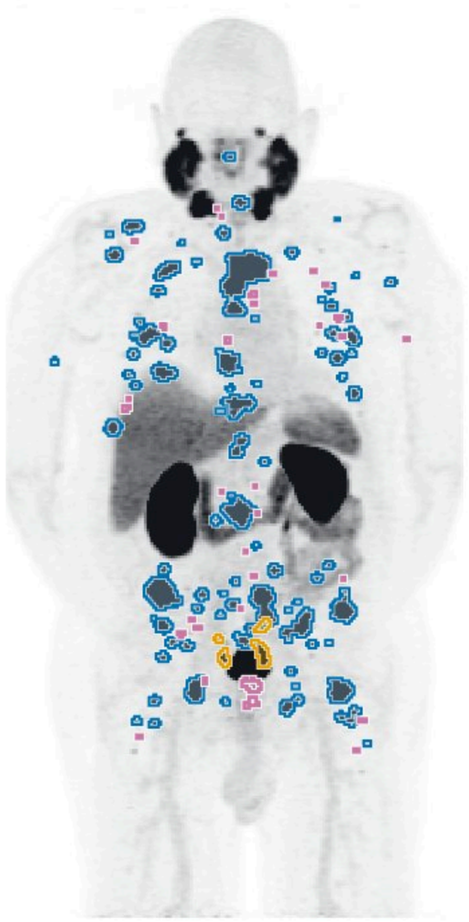
Schematic

F1 balances lesions found against false alarms

Dice shows how well the outlines match

[9] Jeblick et al., Sci Data 2026 · [10] Jackson et al., DEEP-PSMA design document 2025

Results on the Sealed Test Set



Sealed test set, 130 scans

F1 84.3%

82% Found

86% Flagged and real

59.6% Dice
over the 122 scans with lesions

- Found
- Missed
- False alarm

The scan: one sealed DEEP-PSMA test scan, its cohort's median case by F1

autoPET-III, 110 scans

F1	84.3%
Found	85.9%
Flagged and real	82.7%

PSMASegmentator trained on these scans

DEEP-PSMA, 20 scans

F1	84.3%
Found	75.1%
Flagged and real	96.0%

My model trained on 80 of the 100

Neither dataset here gives a fair comparison

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Comparison with PSMASegmentator

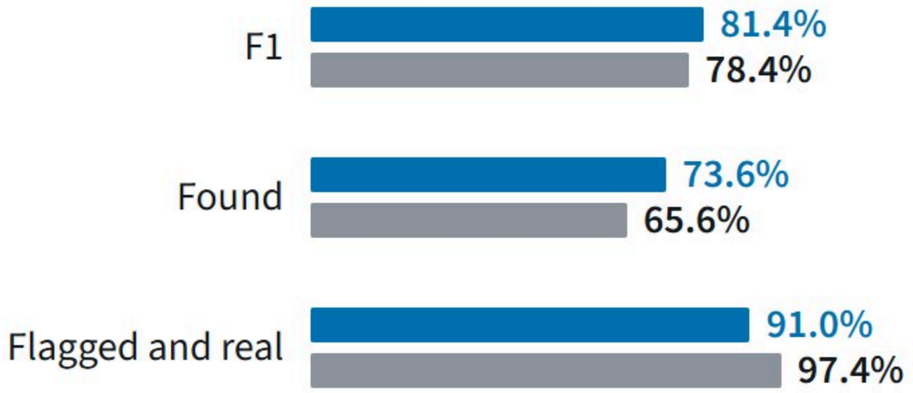
Hybrid CNN-Mamba
a single version,
trained only on autoPET

430 training scans
one model

PSMASegmentator

1,035 training scans
five models, averaged

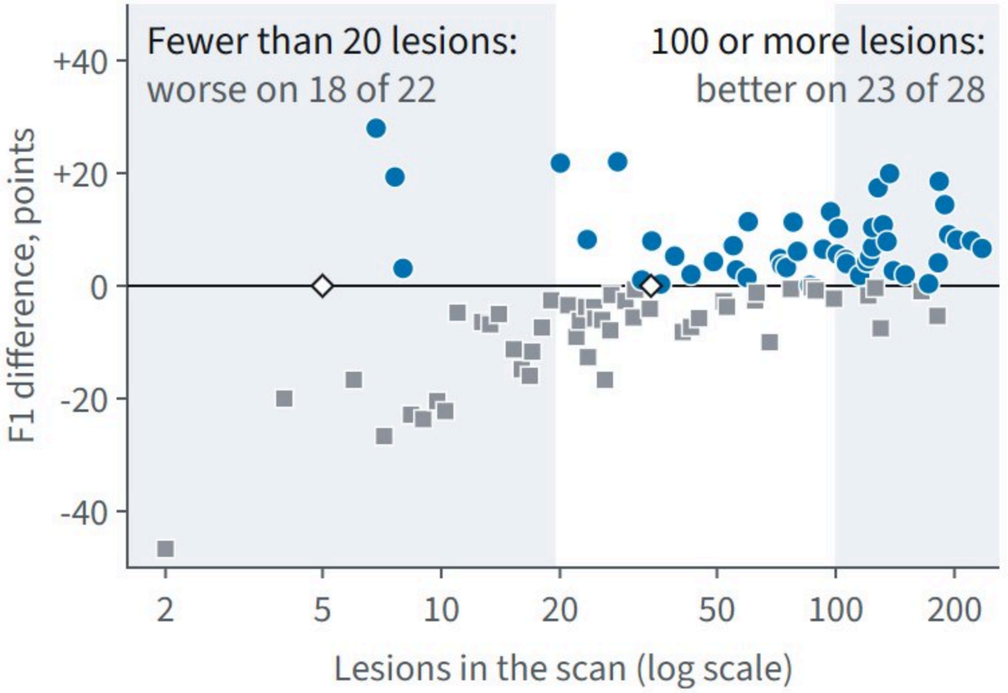
Pooled over 100 DEEP-PSMA scans neither model trained on



+3.0 points
95% interval 1.2 to 4.7

Scan by scan, hybrid minus PSMASegmentator

Better on 48 scans, worse on 50, tied on 2



Mamba Blocks and Patch Size

● Hybrid CNN-Mamba

■ CNN-only version

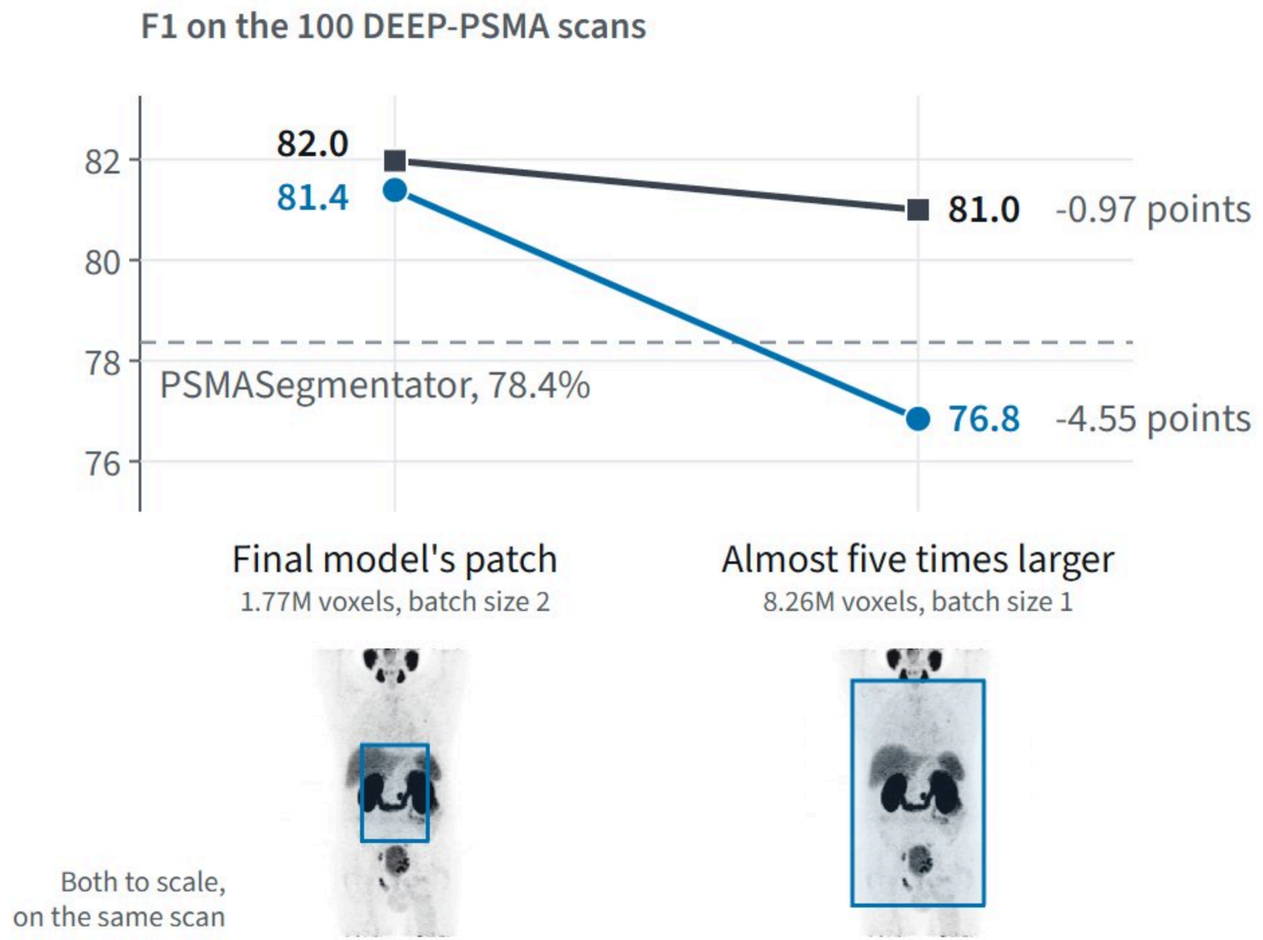
Same network, every Mamba block removed
Both trained only on autoPET

At the final model's patch

No clear difference on DEEP-PSMA
autoPET test set (secondary): hybrid +1.79 points

Gain over PSMASegmentator not credited to Mamba

Each model trained once



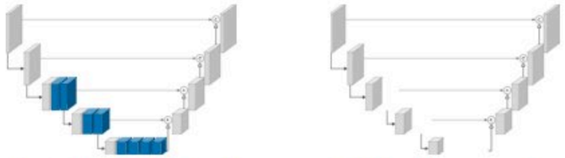
Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Discussion and Limitations

Gain over PSMASegmentator

+3.0 points of pooled F1

95% interval 1.2 to 4.7

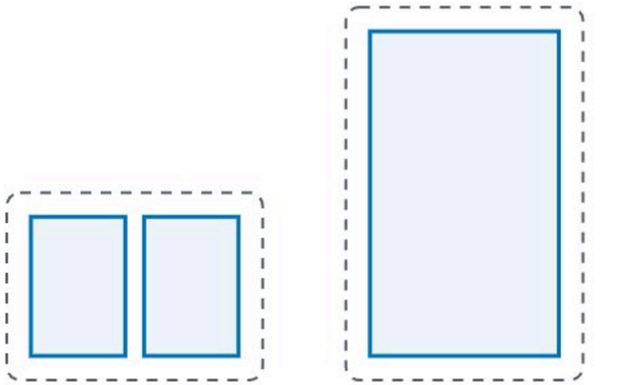


Hybrid CNN-Mamba CNN-only version

Most likely something both networks share, not Mamba

U-Mamba retest, 2024:
no effect from its Mamba layers against a strong CNN

Patch size and batch size



Batch size 2
final model's patch

Batch size 1
almost five times larger

The larger patch still needed batch size 1

Mean time per scan

one RTX 4090, 20 sealed DEEP-PSMA test scans

Hybrid PSMASegmentator

Default settings: 5 against 40 passes

18.1 s

43.8 s

Same number of passes

18.1 s

17.2 s

About the same time on average

Peak memory

5.2 GiB 4.5 GiB

Limits

Each model in the CNN-only comparison trained once
against a spread of about 0.6 F1 points between the final model's five versions

Not like-for-like with PSMASegmentator
most setup differences favour PSMASegmentator

[11] Isensee et al., MICCAI 2024

Conclusion and Next Step

Final model, sealed test set

F1 84.3%

Hybrid CNN-Mamba vs **PSMASegmentator**

Found more lesions, on scans neither model had seen

single version trained only on autoPET: pooled F1 81.4% against 78.4% at the cost of more false alarms

Hybrid CNN-Mamba vs **CNN-only version**

A CNN-only version scored just as well, so the credit doesn't go to Mamba

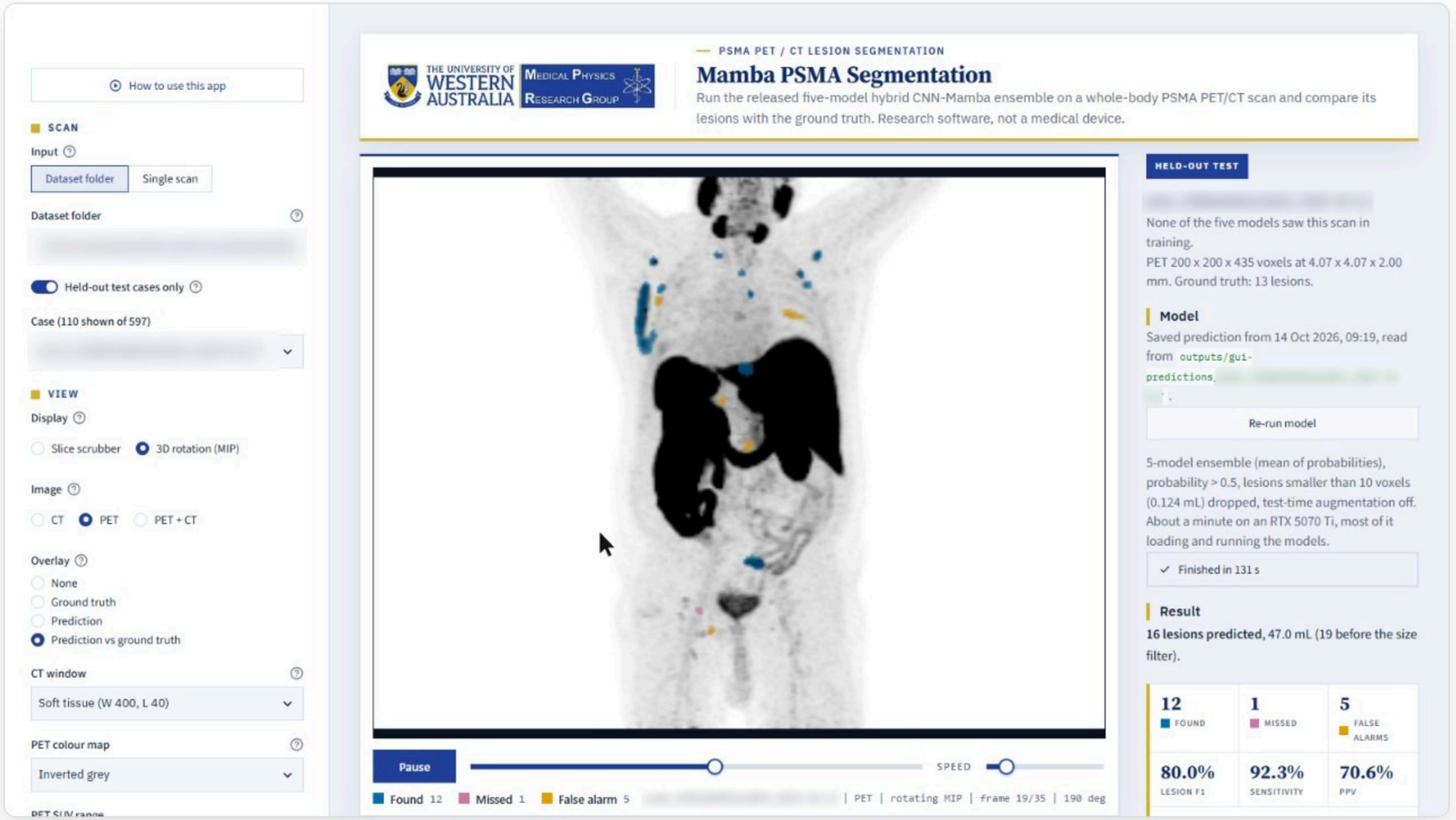
F1 82.0% for the CNN-only version, 81.4% for the hybrid

Thank you

Next step

The code will be released open source, so the UWA Medical Physics Research Group can retrain and test it on Sir Charles Gairdner Hospital data

Recorded Run of the Final Model in the App



My app running the five-model ensemble on a sealed test scan

The result in the app's 3D view

- Found
- Missed
- False alarm

The results slide's colours; before the run, dark blue shows the ground truth.

Waiting for the app is sped up 20x; clicks, scrolling and the 3D turn play in real time.

Threshold and size filter as written down beforehand; no test-time augmentation. The case id is blurred.

Setup Differences with PSMASegmentator

Factor	Hybrid, single version	PSMASegmentator	Favours
Training scans	**430 autoPET-III**	1,035 (597 autoPET-III, 438 Western Australian)	PSMASegmentator
Models at inference	**one**	five, averaged	PSMASegmentator
Test-time augmentation	**off**	on, eight mirrored passes	PSMASegmentator
Deep supervision	**none**	yes	PSMASegmentator
PET/CT misalignment augmentation	**none**	yes	PSMASegmentator
Training patch	**1.77M voxels**	4.72M voxels	PSMASegmentator
Parameters	**39.7M**	about 102.4M	PSMASegmentator
Optimiser	**AdamW, cosine decay**	SGD, polynomial decay	unclear
Post-processing	**none**	disabled for this comparison	neither
Matching rule and scoring grid	**same script and grid**	same script and grid	neither

Most factors favour PSMASegmentator, so the gain is more likely to understate than overstate the recipe; none was tested

Paired Differences on the 100 DEEP-PSMA Scans

Hybrid minus PSMASegmentator	Points	95% interval
Pooled F1	**+3.0**	1.2 to 4.7
Pooled found (sensitivity)	**+8.1**	5.9 to 10.2
Pooled flagged and real (PPV)	**-6.4**	-7.9 to -5.2
Per-scan F1, mean	**-0.6**	-2.8 to 1.6
Per-scan Dice, mean	**+6.4**	4.3 to 8.5

Per scan: better on 48, worse on 50, tied on 2

Fewer than 20 lesions: worse on 18 of 22; 100 or more: better on 23 of 28

Paired bootstrap over patients; pooled F1 weighs lesion-heavy scans the most

False Alarms on Scans with No Lesions

5 of the 8 sealed test scans with no lesions were flagged
by the final five-model ensemble

6 to 8 each version alone, before averaging
the ensemble flagged fewer than any version alone

21 to 7 false alarms on the seven validation scans with no lesions
an earlier version, before and after the organ head

First change to try: the organ head at every deep-supervision level, as PSMASegmentator has

Here the organ head is single-scale, with no post-processing behind it

Lesion Matching Rule and Dice

The rule

Found: a lesion that any predicted voxel touches

False alarm: a predicted component touching no lesion

The same scoring script for both models, on the same grid

written by Joel Noble, used with his permission

The rule is blind to over-segmentation, so Dice is reported beside every lesion figure

Dice on the 100 DEEP-PSMA scans

62.3% Hybrid, single version

55.9% PSMASegmentator

+6.4 points (4.3 to 8.5), consistent with finding more lesions without smearing

Labels of the Two Datasets

autoPET-III

SUV threshold within drawn volumes of interest, then corrected slice by slice by an expert

Sealed test, 110 scans: found 85.9%, flagged and real 82.7%

DEEP-PSMA

SUV of at least 3, physiological uptake removed by a nuclear medicine physician

About 68 separate pieces per scan

Sealed test, 20 scans: found 75.1%, flagged and real 96.0%

Possible explanation, not tested

A model trained mostly on expert-corrected labels covers the bulk of each lesion but misses many small thresholded pieces

Nearly everything it predicts then lands on thresholded disease

The final model trained on 80 of the 100 DEEP-PSMA scans; Dice is not comparable across the two

Mamba Blocks in the Three Deepest Stages

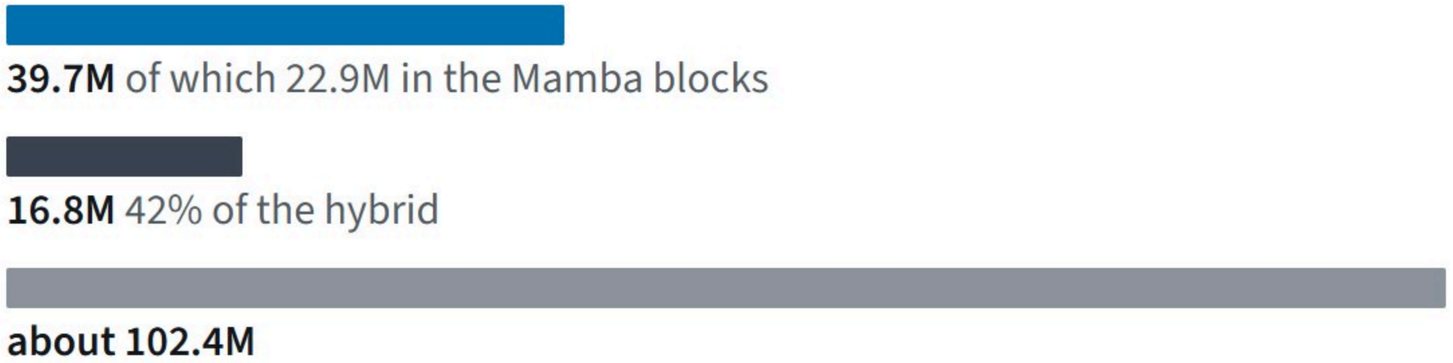
Stage	Channels	Sequence length per direction	Blocks
1	32	1,769,472	convolutions
2	64	221,184	convolutions
3	128	**27,648**	**convolutions and 2 Mamba blocks**
4	256	**3,456**	**convolutions and 2 Mamba blocks**
5	320	**432**	**convolutions and 4 Mamba blocks**

The shallow stages hold low-level texture, which convolution handles more cheaply than a scan of that length

Each Mamba block scans six directions and sums them; neither choice was tested against an alternative

Size of the CNN-Only Version

Parameters



Capacity is a confound in the comparison with PSMASegmentator

In the CNN-only comparison the parameter gap is the same at both patch sizes, so it cannot on its own make the gap between them change with patch size, if the effects add

Comparisons in the CNN-Only Experiment, with Intervals

F1 on the 100 DEEP-PSMA scans	Points	95% interval	Under the rule
Hybrid minus CNN-only, larger patch	**-4.15**	-5.07 to -3.29	real
Hybrid minus CNN-only, final model's patch	**-0.58**	-1.31 to 0.19	no clear effect
Difference between those two gaps	**-3.57**	**-4.72 to -2.48**	**real**
Larger patch, hybrid	**-4.55**	-5.68 to -3.45	real
Larger patch, CNN-only version	**-0.97**	-1.78 to -0.08	no clear effect

Rule, written down before scoring: the interval excludes zero and the size is at least 1.0 point

One training run per model; the final model's five versions spread by about 0.6 points

Larger patch at batch size 1, the final model's patch at 2; several seeds per model, with the larger patch at batch size 2, would settle both

Inference Time in the Dissertation and the Abstract

Mean time per scan, one RTX 4090, 20 sealed DEEP-PSMA test scans	Hybrid	PSMASegmentator
Default settings: 5 against 40 passes	**18.1 s**	43.8 s
Same passes, 5 each	**18.1 s**	17.2 s
Same passes, 40 each	**131.2 s**	43.8 s
Median, 5 passes each	**11.6 s**	17.0 s
Peak memory	**5.2 GiB**	4.5 GiB

The abstract's home-PC benchmark, one RTX 5070 Ti (not in the dissertation)	Hybrid	PSMASegmentator
Published settings, mean seconds per scan	**32**	63
Same passes, 5 each, mean seconds	**32**	33

The default gap comes from PSMASegmentator's test-time augmentation, not the architecture

Cross-Validation and the Five Folds

567 scans in the cross-validation pool, in five folds split by patient
67 in the first fold, 125 in each of the others

81.8% pooled F1 over the 567 scans, each scored by the version that never trained on it
fold-to-fold standard deviation 1.5 points

84.3% F1 of the five versions averaged, on the sealed test set
one version alone: 83.1 on average (standard deviation 0.6)

Test-Set Results by Dataset with Intervals

Sealed test set	F1	Found	Flagged and real
Pooled, 130 scans	**84.3%** 81.4 to 86.5	**82.4%** 77.4 to 86.5	**86.2%** 83.4 to 88.9
autoPET-III, 110 scans	**84.3%** 80.2 to 86.8	85.9% 79.3 to 90.3	82.7% 78.8 to 85.9
DEEP-PSMA, 20 scans (in-domain)	**84.3%** 79.8 to 87.9	75.1% 69.2 to 80.3	96.0% 93.3 to 98.0

95% intervals from a bootstrap over patients, added after the test set was scored; no point estimate changed

In-domain: the final model trained on 80 of the 100 DEEP-PSMA scans

References

[1] J. Kendrick, R. J. Francis, G. M. Hassan, P. Rowshanfarzad, J. S. L. Ong, and M. A. Ebert, "Fully automatic prognostic biomarker extraction from metastatic prostate lesion segmentations in whole-body [68Ga]Ga-PSMA-11 PET/CT images," European Journal of Nuclear Medicine and Molecular Imaging, vol. 50, pp. 67-79, 2022.

[2] S. Sheikhbahaei, A. Afshar-Oromieh, M. Eiber et al., "Pearls and pitfalls in clinical interpretation of prostate-specific membrane antigen (PSMA)-targeted PET imaging," European Journal of Nuclear Medicine and Molecular Imaging, vol. 44, pp. 2117-2136, 2017.

[3] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," Nature Methods, vol. 18, no. 2, pp. 203-211, 2021.

[4] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv preprint arXiv:2312.00752, 2023.

[5] Z. Xing, T. Ye, Y. Yang, D. Cai, B. Gai, X.-J. Wu, F. Gao, and L. Zhu, "SegMamba-V2: Long-range sequential modeling Mamba for general 3-D medical image segmentation," IEEE Transactions on Medical Imaging, vol. 45, no. 1, pp. 4-15, 2026.

[6] Z. Xing, T. Ye, Y. Yang, G. Liu, and L. Zhu, "SegMamba: Long-range sequential modeling Mamba for 3D medical image segmentation," in Medical Image Computing and Computer Assisted Intervention (MICCAI 2024), 2024.

[7] M. Rokuss, B. Kovacs, Y. Kirchhoff, S. Xiao, C. Ulrich, K. H. Maier-Hein, and F. Isensee, "From FDG to PSMA: A hitchhiker's guide to multitracer, multicenter lesion segmentation in PET/CT imaging," arXiv preprint arXiv:2409.09478, 2024.

[8] J. Wasserthal, H.-C. Breit, M. T. Meyer, M. Pradella, D. Hinck, A. W. Sauter, T. Heye, D. T. Boll, J. Cyriac, S. Yang, M. Bach, and M. Segeroth, "TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images," Radiology: Artificial Intelligence, vol. 5, no. 5, p. e230024, 2023.

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